

INCITE Report and Renewal Contribution

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1 Project Achievements

1.1 Significant Accomplishments to Date

The INCITE allocation helped make advances in theoretical and computational methods used to study self-trapped excitons (STEs) from first-principles calculations. STEs have been shown to cause broadband white light emission in certain double perovskites, showing coupling to phonons, Stokes shifts, and broad emission peaks [1, 2]. Being able to explain this type of emission from first-principles calculations would help connect the measured spectra to lattice distortions, phonon modes, and exciton-phonon coupling parameters.

1.1.1 Previous Methods and Challenges

There are several ways to study STEs, but each method has either computational challenges or methodological limitations. The methods in the literature can be grouped into three main approaches:

- Restricted Open-Shell Kohn-Sham method [3].
- Excited-state forces [4, 5].
- Minimization under the harmonic lattice approximation [6, 7].

ROKS has been shown to describe STE states from density-functional theory (DFT) [3]. This can describe an excited-state structure by fixing occupations, but it does not explain all effects that come from electron-hole correlations in excitons.

Other methods account for electron-hole correlations by using excitons from the Bethe-Salpeter Equation (BSE) under the GW approximation. Excited-state forces (ESF) compute forces from these excitons and use them to relax the system into the STE state [4, 5]. ESF can capture the large lattice distortion that forms an STE, including non-harmonic effects. However, the calculation is expensive, especially in supercells, and it can depend strongly on the initial lattice displacement used in the relaxation process. A third approach is to minimize the exciton energy under a harmonic lattice approximation [6, 7]. This is computationally less expensive as it does not require a supercell, but it misses non-harmonic lattice distortions.

One major limitation of all these methods is that they do not provide a way to calculate the temperature dependence of STE properties, which has been observed experimentally in $\text{Cs}_2\text{AgInCl}_6$ [8].

In this project, we made methodological developments and addressed computational challenges based on these methods in the literature. The work can be roughly grouped into two main points:

- Developing a many-body perturbation theory description of STEs to explain temperature-dependent effects. This includes approximating the energies using tadpole and Fan Migdal diagrams, together with code development with partial self-consistency to obtain STE states under the harmonic approximation.
- Reducing the computational cost of ESF by using a better initial guess from the self-energy method.

1.1.2 Self-Energy Method Development

We developed partially self-consistent self-energy formulations to describe STE states, extending prior treatments of exciton-phonon coupling and excitonic polarons [9, 6, 7]. The tadpole and Fan Migdal diagrams give a Dyson-like equation for the STE states, which is then solved with partial self-consistency.

The tadpole diagram captures the static lattice distortion driven by the exciton. The Fan Migdal term gives access to temperature-dependent effects that minimization methods miss. Together, within the harmonic approximation, they yield the phonon modes contributing to the STE emission peak and the temperature dependence of the peak width.

Theory and code development consumed a large part of the allocation. Both self-energy terms had to be implemented, debugged, and tested. Only the lowest eigenvalues and eigenvectors of the self-energy matrix were computed, targeting the low-energy STE states. Parallelization details are discussed in Section 1.3.

The method was first benchmarked on LiF and SiO_2 , which have prior STE results in the literature and, for SiO_2 , experimental data. The quantities used for comparison included the STE energy, the projection of the STE wavefunction onto exciton and phonon band states, and the real-space localization of the wavefunction. These benchmarks identified the phonon modes most responsible for STE formation and showed how the exciton localizes in the STE state. The method was then applied to $\text{Cs}_2\text{AgInCl}_6$. Compared to LiF, $\text{Cs}_2\text{AgInCl}_6$ shows a stronger Fan Migdal contribution, driven by a lower-energy optical mode that couples more strongly with the exciton: the relevant phonon mode is around 15 meV in $\text{Cs}_2\text{AgInCl}_6$, compared to around 60 meV in LiF.

Finite-momentum exciton calculations were carried out for all three materials, LiF, SiO_2 , and $\text{Cs}_2\text{AgInCl}_6$, on $2 \times 2 \times 2$, $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grids, using Quantum ESPRESSO for

DFT, DFPT, and EPW calculations, and BerkeleyGW for GW and GW-BSE calculations. These inputs were needed at each grid size to extrapolate convergence toward an infinite grid limit.

1.1.3 Excited-State Forces and STE Relaxation

The ESF method was introduced by Ismail-Beigi and Louie [4] and further developed by Del Grande and Strubbe [5]. ESF calculations on LiF and SiO₂ were used to validate the workflow against prior literature results. For Cs₂AgInCl₆, however, convergence difficulties arose: some starting distortions led to slow convergence or oscillations. Since each ESF step requires expensive electronic-structure work, repeatedly trying different starting distortions is not practical. The self-energy calculation gives a better initial guess, which reduced the number of ESF iterations needed and made the calculation feasible on Frontier.

Each ESF iteration for Cs₂AgInCl₆ involved a full sequence of DFT, DFPT, EPW, GW, and BSE calculations on a $3 \times 3 \times 3$ supercell. The initial lattice distortion for the first step was taken directly from the self-energy calculation, ensuring a starting point tied to the STE rather than an arbitrary displacement. At each subsequent step, the distortion is updated from the forces computed in the previous iteration.

The quantities extracted from the ESF calculation are the total energy change relative to the undistorted structure and the final electronic wavefunction, which is compared against the exciton state to assess how much the STE state resembles the original exciton and how it has changed due to the lattice relaxation.

1.1.4 Main Accomplishments and Impact

The main work completed to date is:

- A partially self-consistent Dyson-like equation with tadpole and Fan Migdal contributions was derived and implemented for predicting STE states under the harmonic approximation.
- The code was used to calculate STE states in LiF, SiO₂, and Cs₂AgInCl₆.
- The self-energy distortion was used as the initial guess for ESF calculations on Cs₂AgInCl₆, improving convergence and reducing computational cost.
- Properties including the Stokes shift, temperature dependence of the emission peak width, real-space STE wavefunction, and projection of the STE onto exciton and phonon components were computed to characterize STE behavior.

The methods developed here are general and can be applied to study STE states in a wide range of materials, including two-dimensional systems and Ruddlesden-Popper phases, where exciton-phonon coupling and self-trapping are also of interest.

1.2 Allocation Use

1.2.1 Usage to Date

About 1,270 node hours have been used to date on Frontier by user `krishnaa423`. Initially a lot of time was spent in code development, debugging, and close analysis of intermediate results before running larger calculations. Table 1 shows the approximate breakdown by activity.

Activity	Node-Hour %	Systems
Self-energy code development, debugging, testing	~20%	LiF (primary test case)
Finite-momentum exciton input generation	~30%	LiF, SiO ₂ , Cs ₂ AgInCl ₆
ESF iterations (DFT, DFPT, EPW, GW, BSE cycles)	~50%	Cs ₂ AgInCl ₆

Table 1: Approximate node-hour breakdown on Frontier.

- **Self-energy code development, debugging, and testing (~20%):** Initial implementation and validation of the self-energy code was done primarily on LiF, including debugging, benchmarking against literature results, and testing parallelization.
- **Finite-momentum exciton input generation (~30%):** DFPT and EPW calculations via Quantum ESPRESSO, followed by GW-BSE calculations via BerkeleyGW, were run for LiF, SiO₂, and Cs₂AgInCl₆ on $2 \times 2 \times 2$, $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grids. These inputs feed directly into the self-energy calculations.
- **ESF iterations (~50%):** The dominant cost was the ESF calculation on Cs₂AgInCl₆, which required approximately six full DFT, DFPT, EPW, GW, and BSE cycles. Each iteration is expensive because it involves a complete electronic-structure and BSE calculation at each step.

1.2.2 Projected Use

For the remainder of the current allocation year, approximately 2,000 node hours are anticipated for computing STE properties on the current systems, LiF, SiO₂, and Cs₂AgInCl₆, along with associated code testing and debugging. This includes quantities such as exciton projections, phonon contributions, real-space wavefunctions, and temperature-dependent linewidths from both methods. Following this, time will be spent on manuscript preparation before pursuing calculations on new materials such as two-dimensional systems or Ruddlesden-Popper phases. Usage will remain intermittent rather than evenly spread.

1.3 Application Parallel Performance

The production work used established electronic-structure codes alongside the in-house self-energy code. Quantum ESPRESSO and EPW handled DFT, DFPT, and electron-phonon inputs. BerkeleyGW was used for GW and finite-momentum BSE calculations. BerkeleyGW parallelizes well over Frontier: the finite-momentum calculations over $\mathbf{Q}$ points were split into independent batches, and the GW and BSE stages were parallelized over bands and $\mathbf{k}$ points.

The core computational step in the self-energy code is solving a Dyson-like equation whose lowest-energy solutions give the STE eigenvalues and eigenvectors. Rather than implement a custom parallel eigensolver, we used PETSc and SLEPc through the Python interfaces `petsc4py` and `slepc4py`, all compiled with HIP support for Frontier’s AMD GPUs. This provides distributed arrays and matrices across GPUs and gives access to parallel eigensolvers that target only the lowest part of the spectrum. This way the calculation is effectively parallelized over Q points, which made the $4 \times 4 \times 4$, $6 \times 6 \times 6$, and $8 \times 8 \times 8$ grid runs more practical. The resulting wavefunctions are written to `.h5` files via `h5py` for parallel I/O.

The self-energy result also provided a better initial guess for the lattice distortion used in the excited-state force calculations, reducing the number of lattice displacement steps needed to reach the relaxed STE geometry.

The main remaining performance challenge is DFPT. A Quantum ESPRESSO build that uses Frontier’s AMD GPUs effectively for this part of the workflow has not been identified, so input

generation for the electron-phonon stages remains less accelerated than the BerkeleyGW and self-energy stages.

2 Project Plans for Next Year

2.1 Summary of Project Plans

The next allocation period will focus on analyzing the completed calculations and preparing manuscripts before starting any new calculations. The goal is to extract STE observables from the data already in hand: the relaxed lattice distortion, the Stokes shift, the emission linewidth, and the temperature dependence of the photoluminescence spectrum.

The first priority is extracting and comparing properties from all completed calculations across all systems and both methods. This means computing the STE wavefunction, exciton projections onto band states, phonon-resolved contributions, Stokes shifts, and temperature-dependent emission linewidths for LiF, SiO₂, and Cs₂AgInCl₆, using both the self-energy and ESF approaches wherever both are available. Comparing the two methods across all three systems will show when the harmonic self-energy picture is sufficient and when non-harmonic relaxation changes the result. The goal at this stage is to have enough processed output to read and understand the data, before any writing begins.

The second priority is manuscript preparation. Two papers are planned: one on the self-energy framework for STEs and its temperature-dependent predictions, and one on the Cs₂AgInCl₆ STE using the self-energy-guided ESF result.

Extension to new materials such as two-dimensional systems or Ruddlesden-Popper phases will be considered only after the current results are finalized and the manuscripts are underway. Based on current usage, a new material requiring finite-momentum exciton inputs and self-energy calculations is estimated at roughly 1,000 node hours. Adding ESF iterations would bring the total to roughly 2,000 node hours per material. These estimates could increase significantly depending on the system size, since larger unit cells raise the cost of the GW, BSE, and DFPT stages, and two-dimensional systems in particular may require methodological development to parallelize well over Frontier.

2.2 Development Work

Development during the current allocation period focused on the theory and implementation of the STE self-energy method, including the self-consistent tadpole term, the Fan Migdal term, finite-momentum exciton inputs, and post-processing for STE observables. Development also included tests on LiF, debugging, extension to denser grids, and the Frontier build of PETSc, SLEPc, `petsc4py`, and `slepc4py` with HIP support for AMD GPUs.

Future development work will focus on the post-processing code needed to extract STE properties from completed calculations, including exciton projections, phonon-resolved contributions, real-space wavefunctions, and temperature-dependent linewidths. Some additional parallelization optimization may also be needed depending on new systems and higher grid sizes.

3 Publications Resulting from INCITE Awards

There are no submitted or published papers yet from this INCITE work that include the required INCITE/OLCF acknowledgement. Two manuscripts are in preparation from the calculations completed so far:

- A paper on the first-principles exciton self-energy framework for STEs, including temperature-dependent effects and photoluminescence linewidths.
- A paper on $\text{Cs}_2\text{AgInCl}_6$, using self-energy-guided excited-state forces to obtain the STE distortion, Stokes shift, and emission properties of the double perovskite.

These papers will include the required INCITE/OLCF acknowledgements when submitted.

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